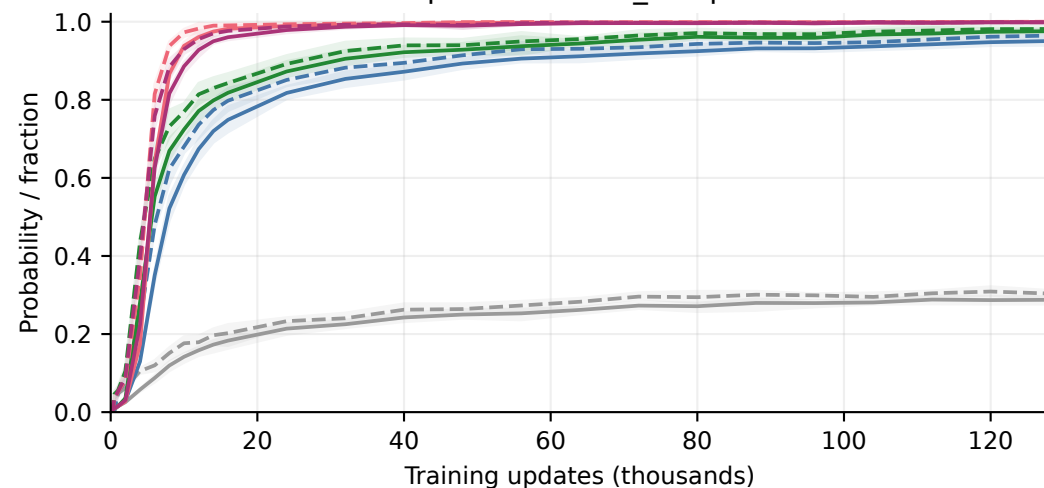
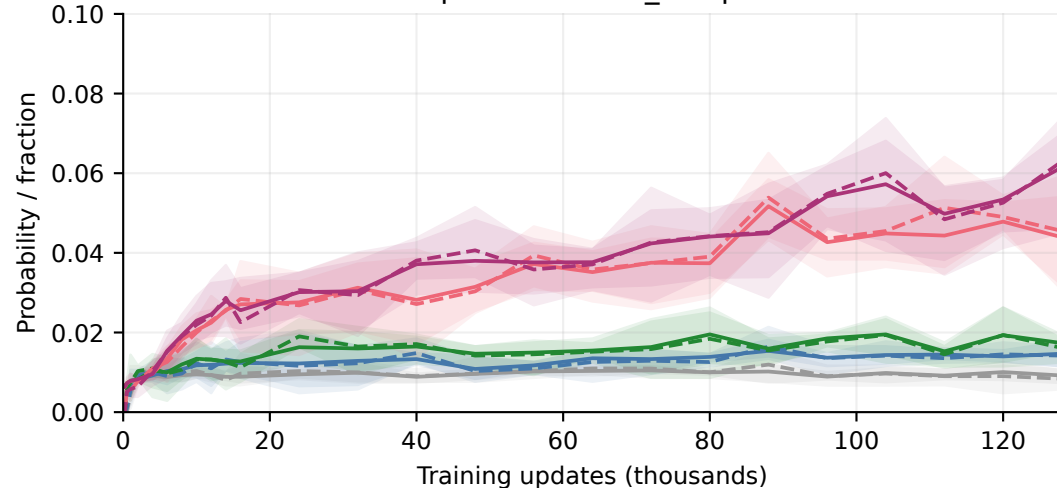


# Confirmation: composition; bands span world means

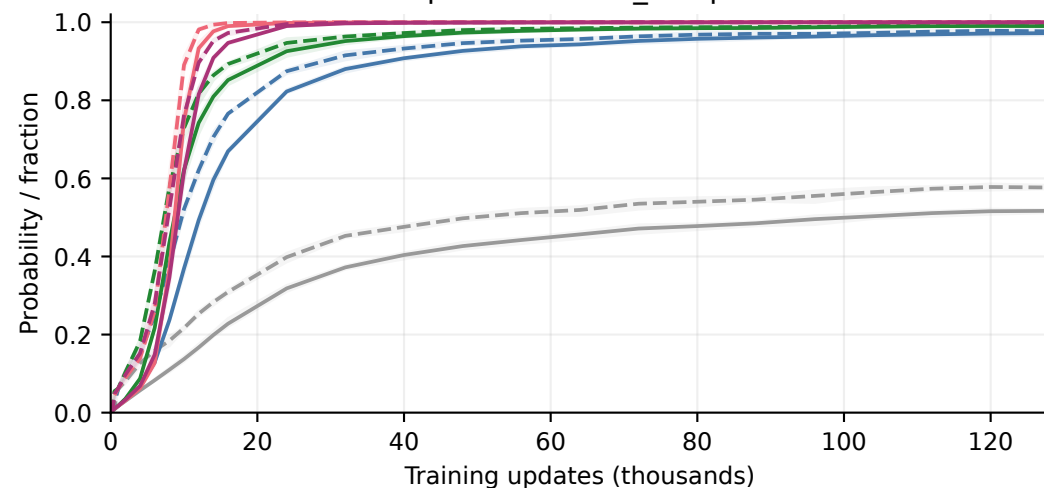
## 2-hop models: test\_composite



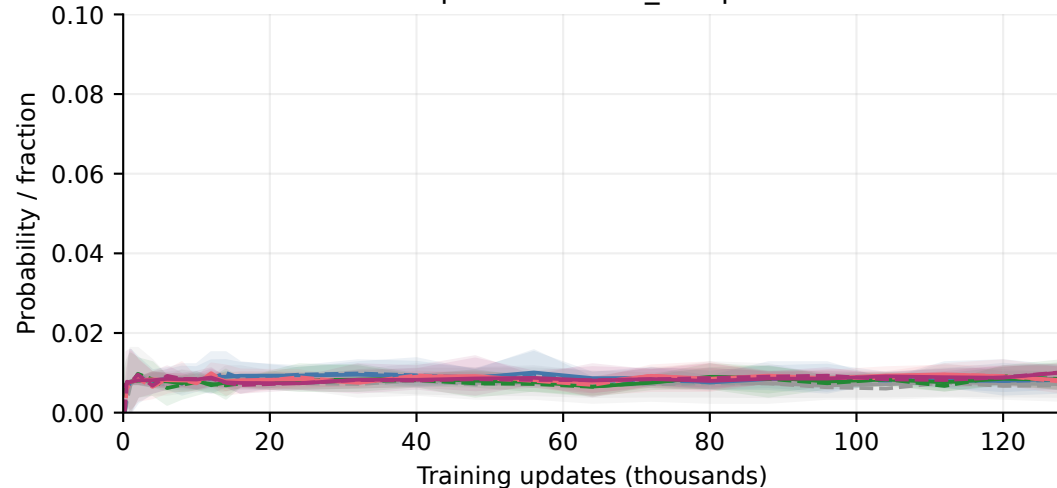
## 2-hop models: ood\_composite



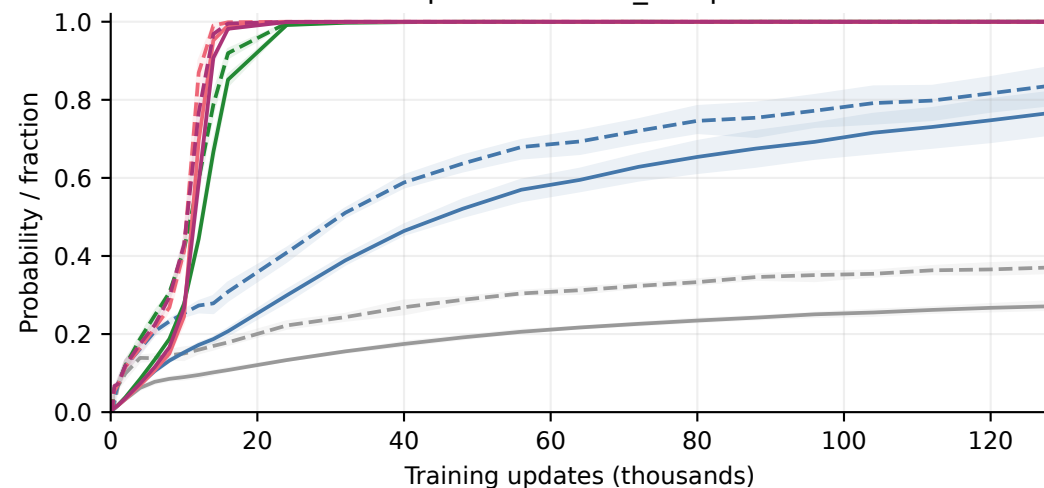
## 3-hop models: test\_composite



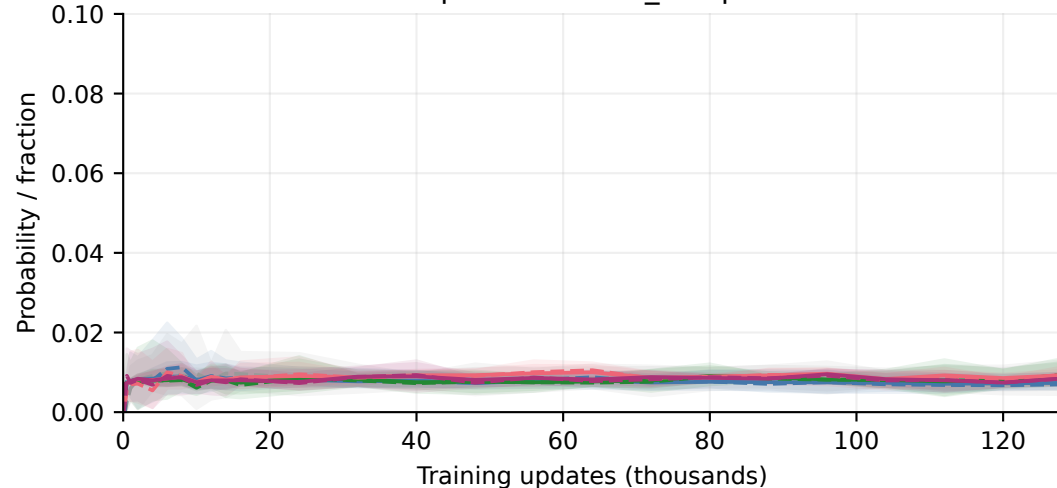
## 3-hop models: ood\_composite



## 4-hop models: test\_composite



## 4-hop models: ood\_composite

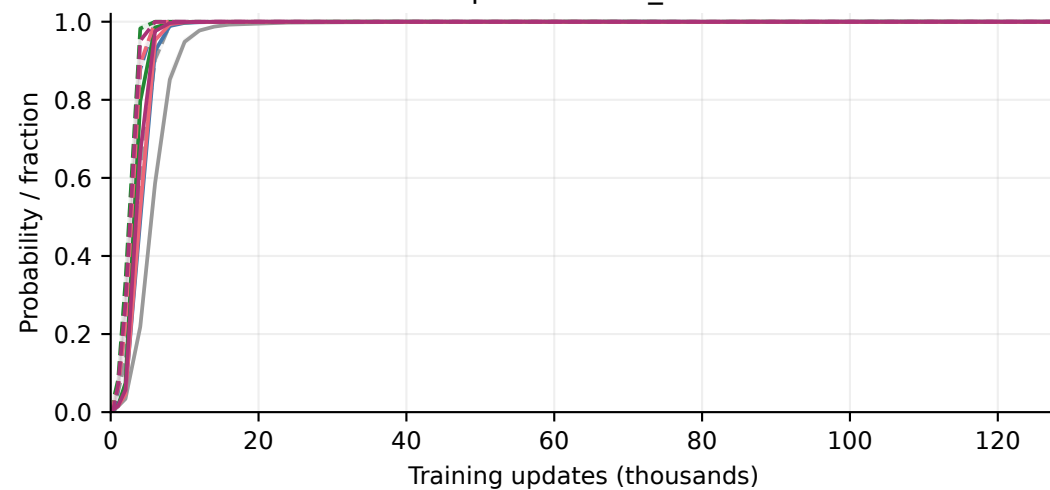


C1    CD    L2  
 C2    L1    Mean P(gold answer)

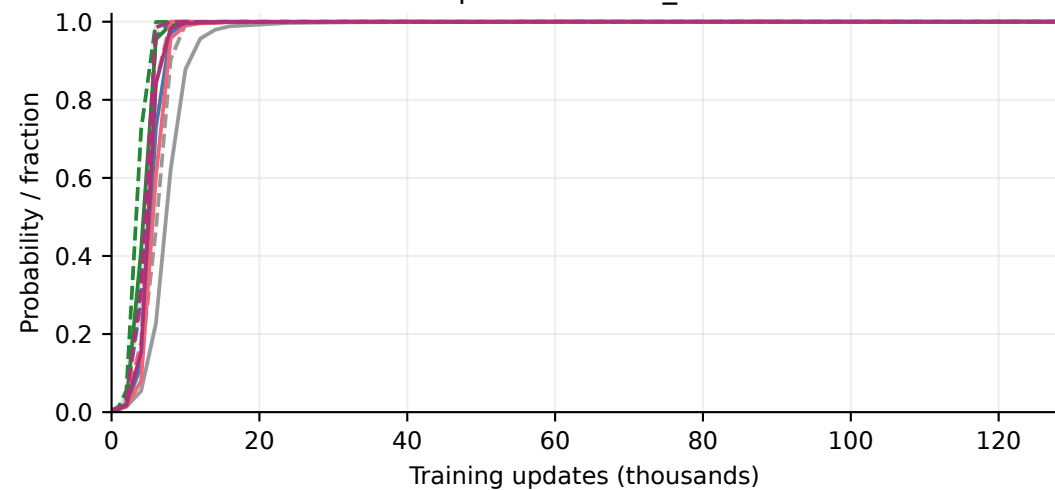
--- Answer accuracy

# Confirmation: atomic facts; both classes trained as single hops

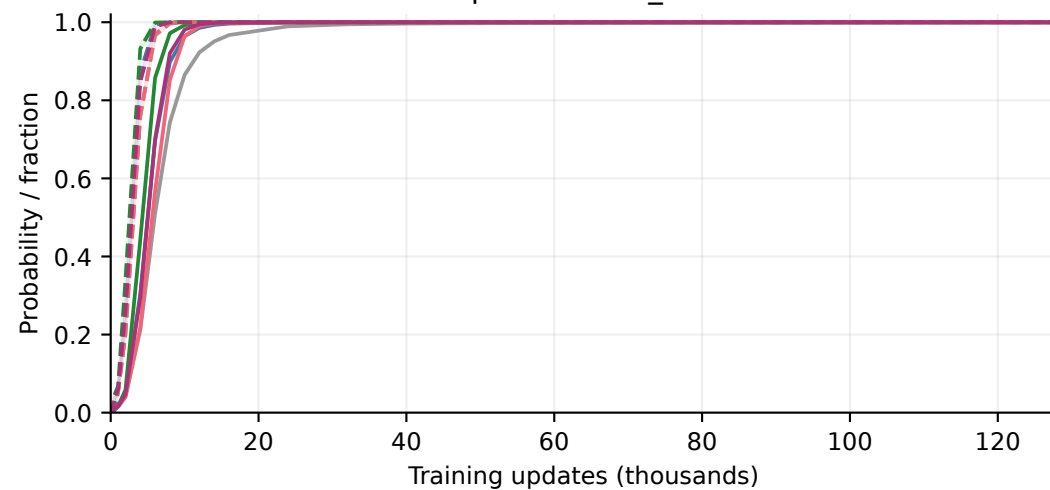
## 2-hop models: id\_atomic



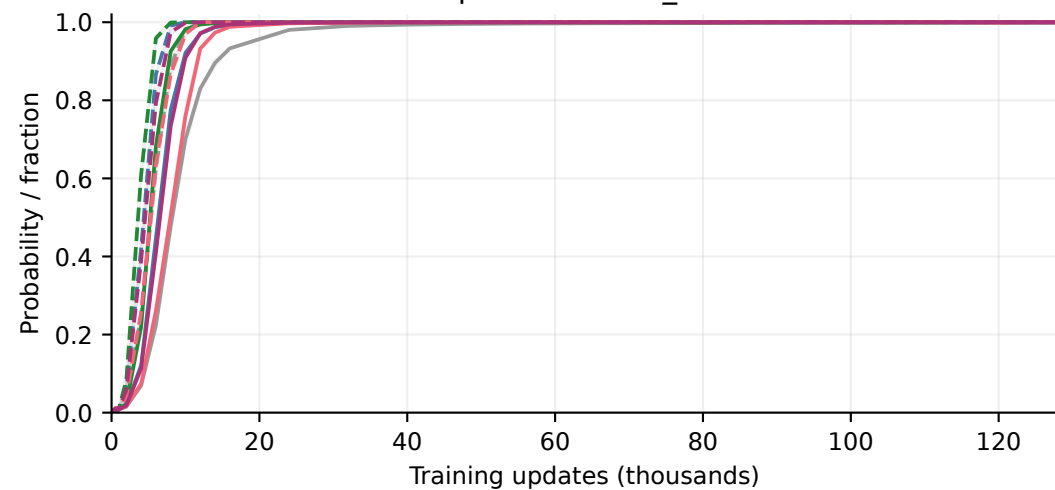
## 2-hop models: ood\_atomic



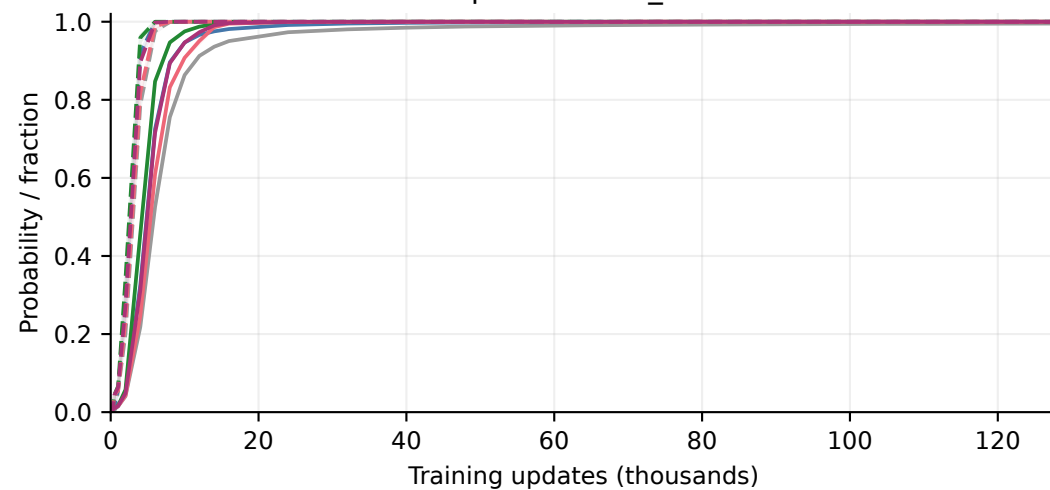
## 3-hop models: id\_atomic



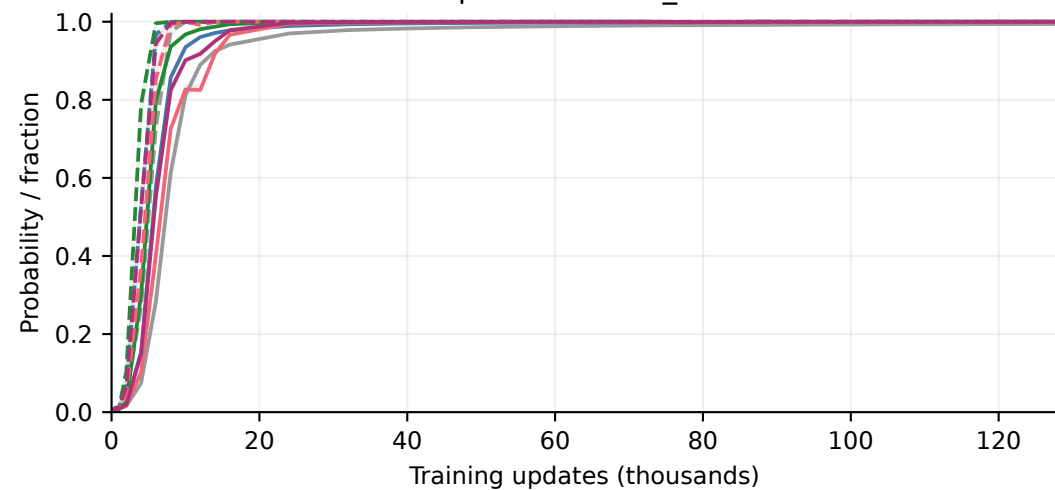
## 3-hop models: ood\_atomic



## 4-hop models: id\_atomic



## 4-hop models: ood\_atomic

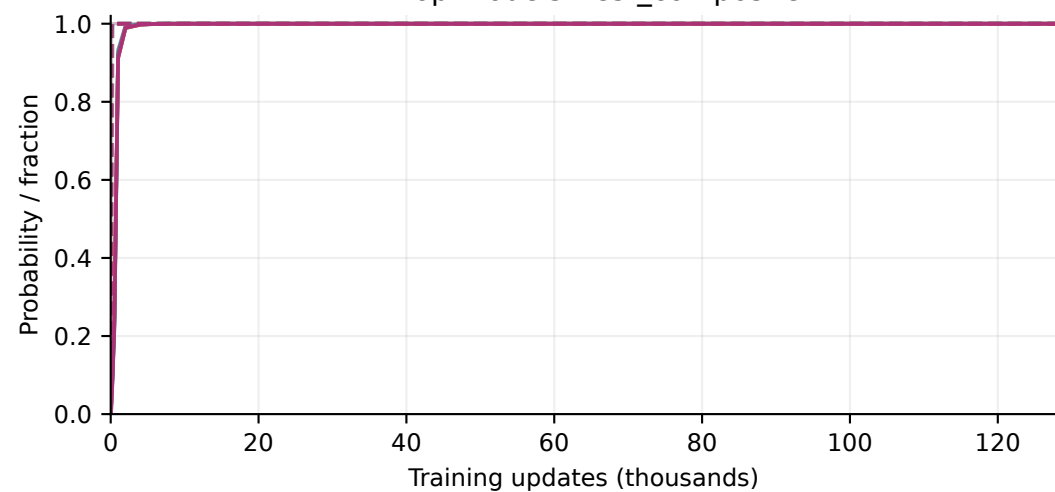


C1   
  CD   
  L2  
 C2   
  L1   
  Mean P(gold answer)

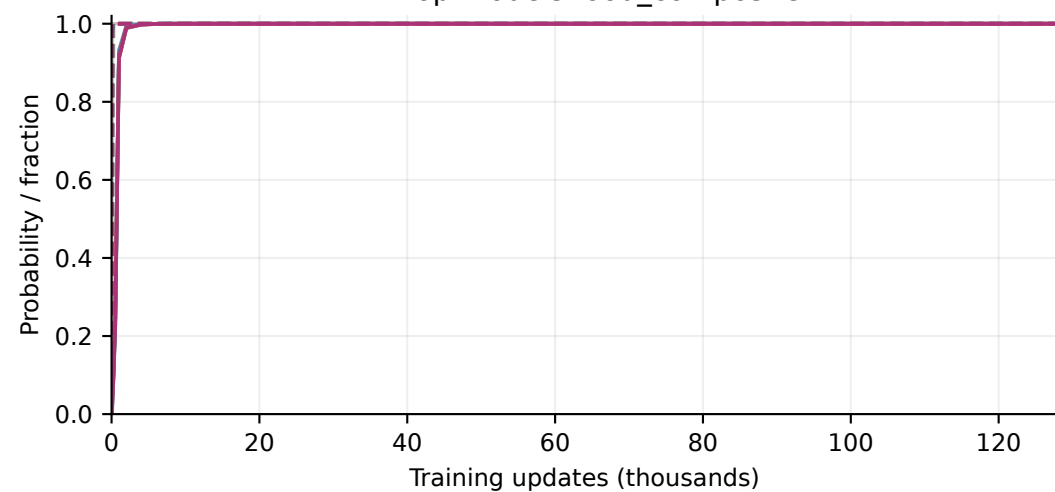
Answer accuracy

## EOS: gold-answer conditioning and generated-answer stopping are different

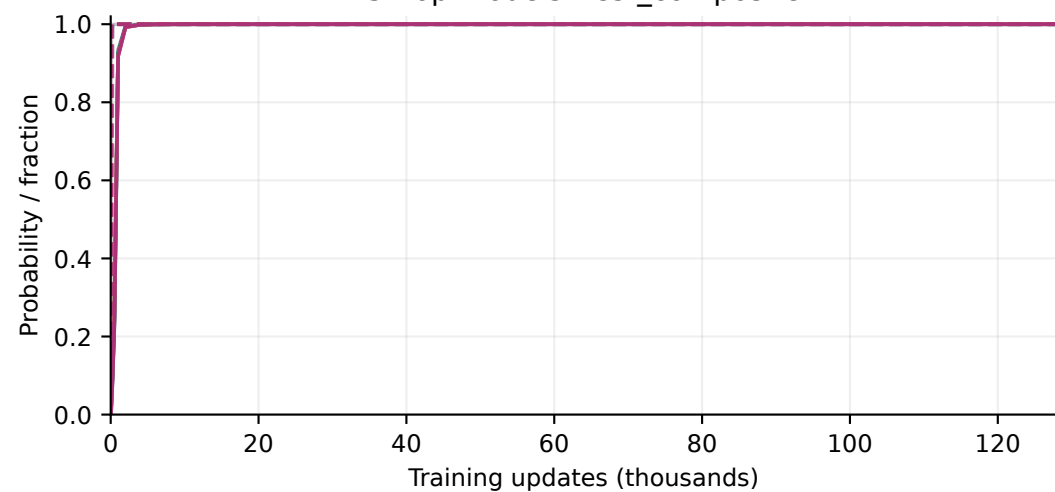
2-hop models: test\_composite



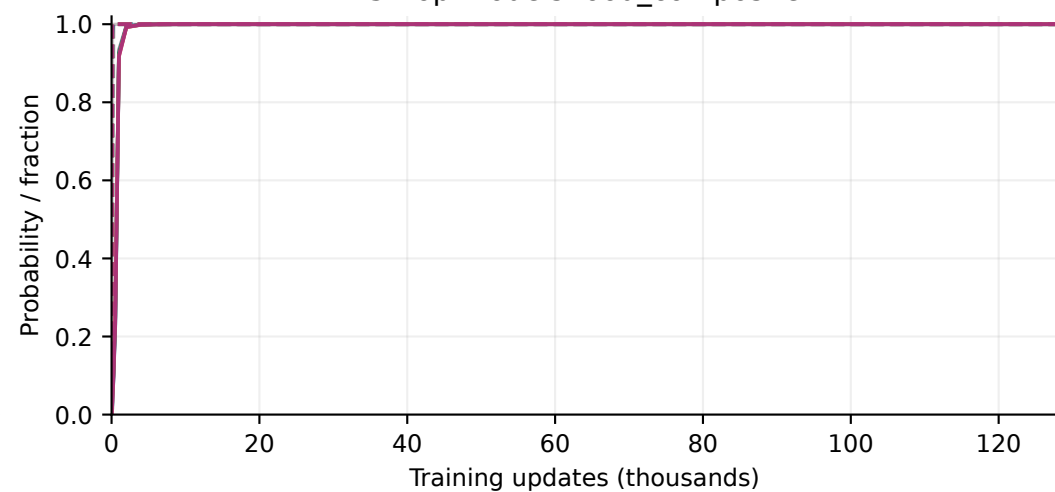
2-hop models: ood\_composite



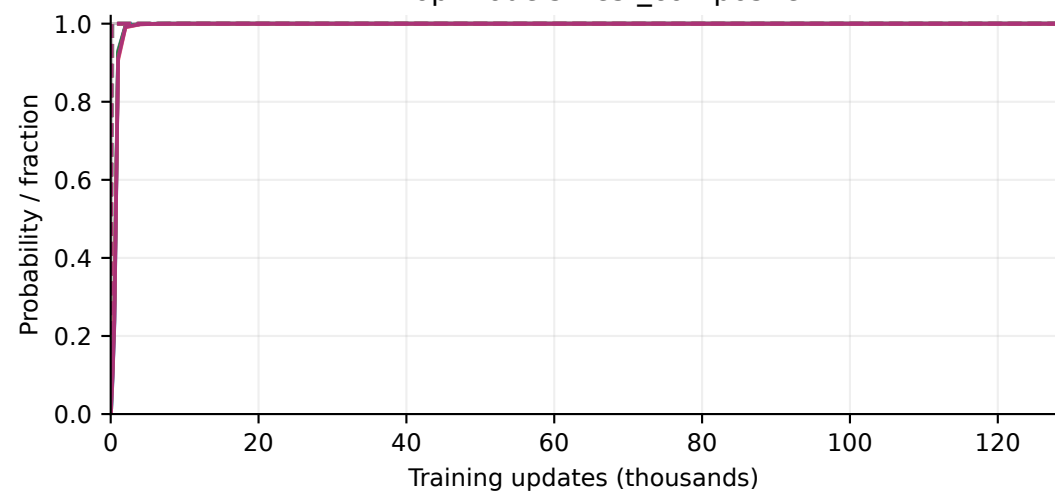
3-hop models: test\_composite



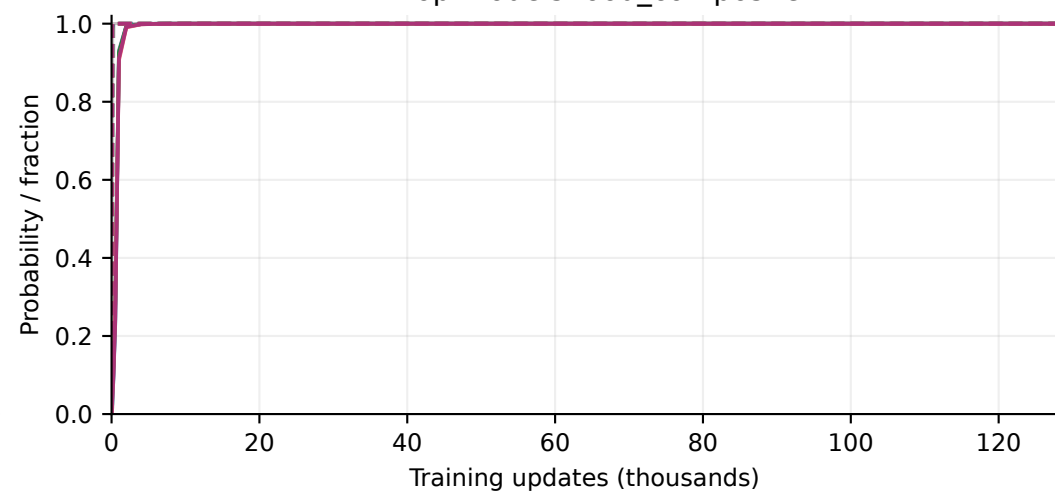
3-hop models: ood\_composite



4-hop models: test\_composite



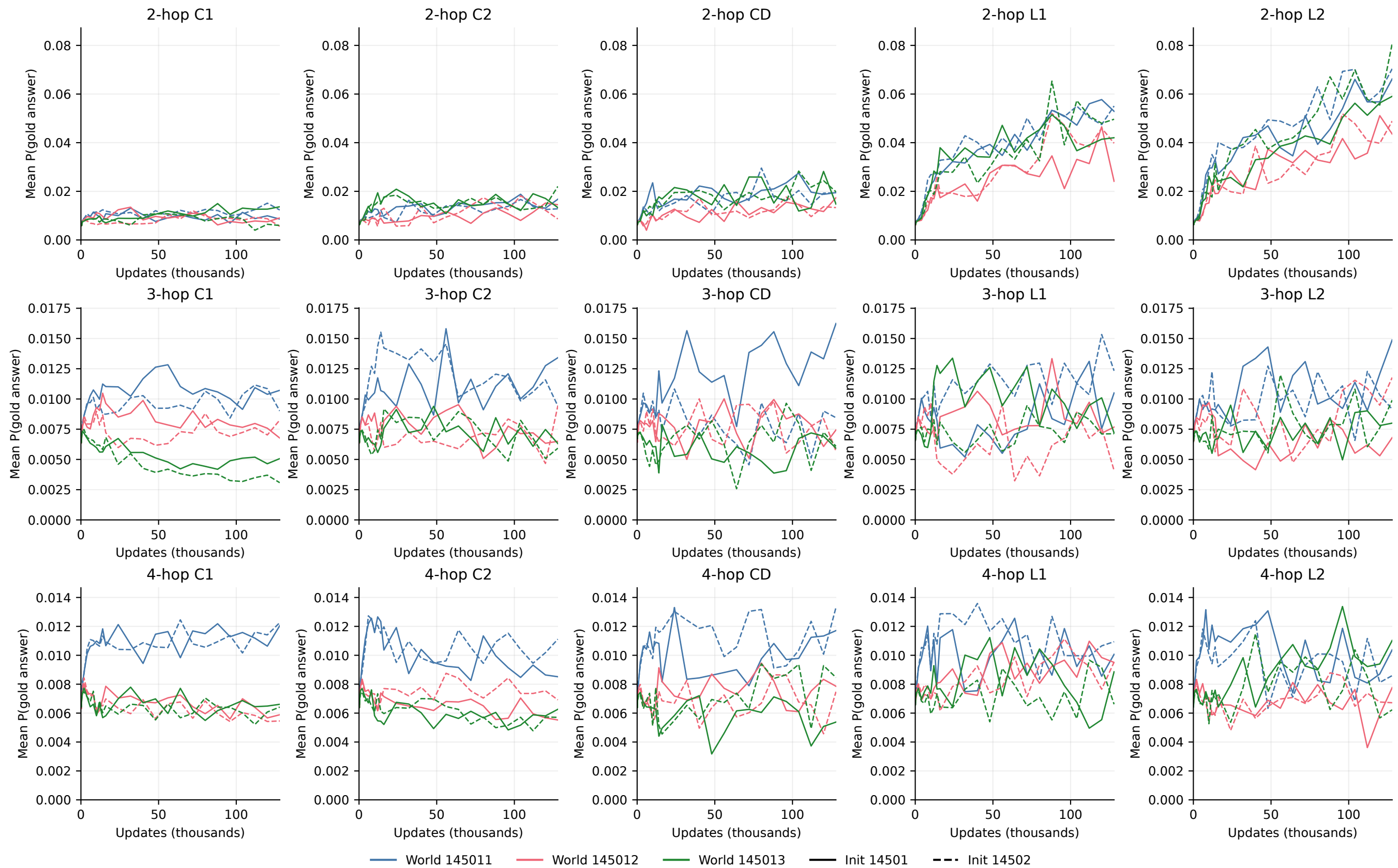
4-hop models: ood\_composite



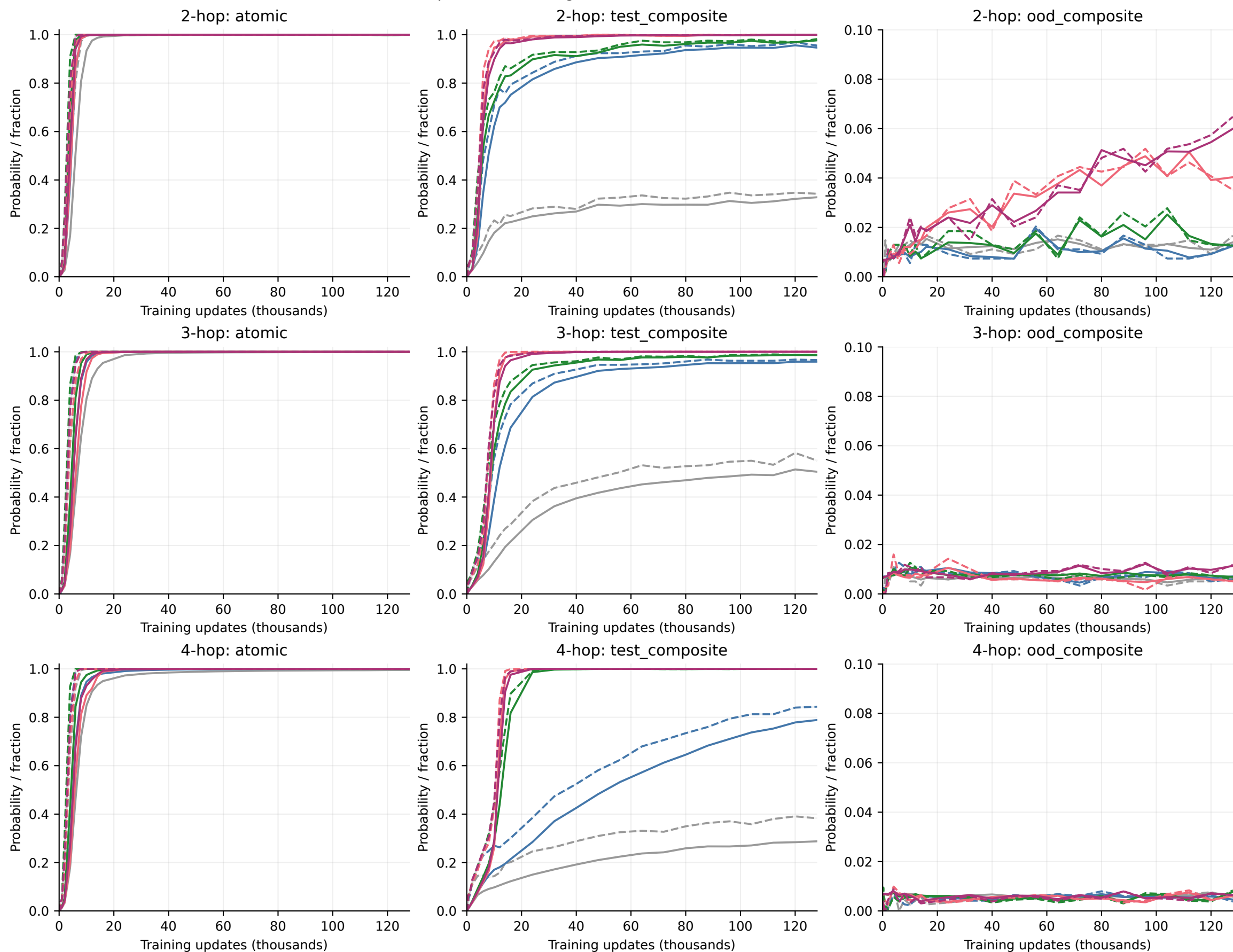
C1 CD L2  
C2 L1 P(EOS | gold answer)

Generated EOS accuracy

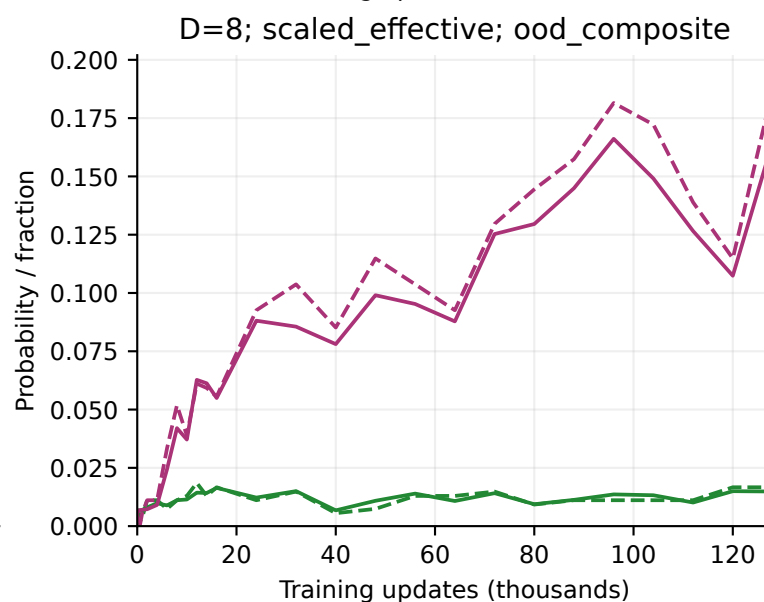
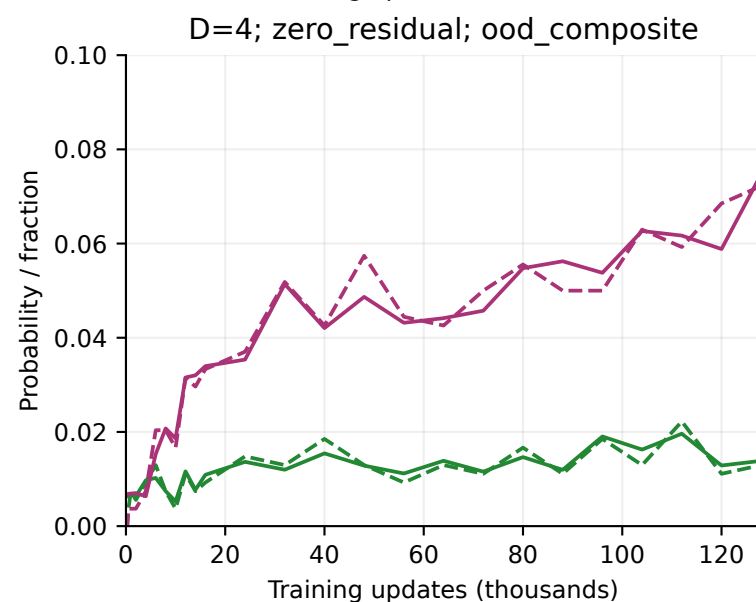
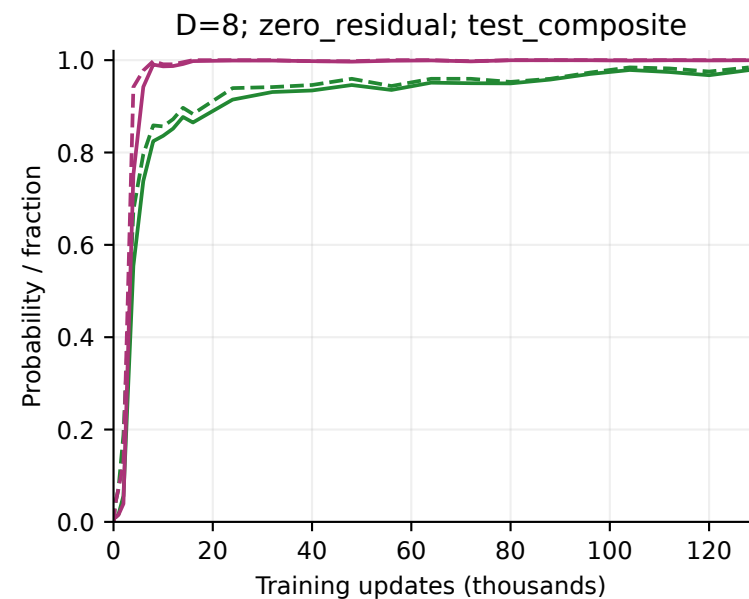
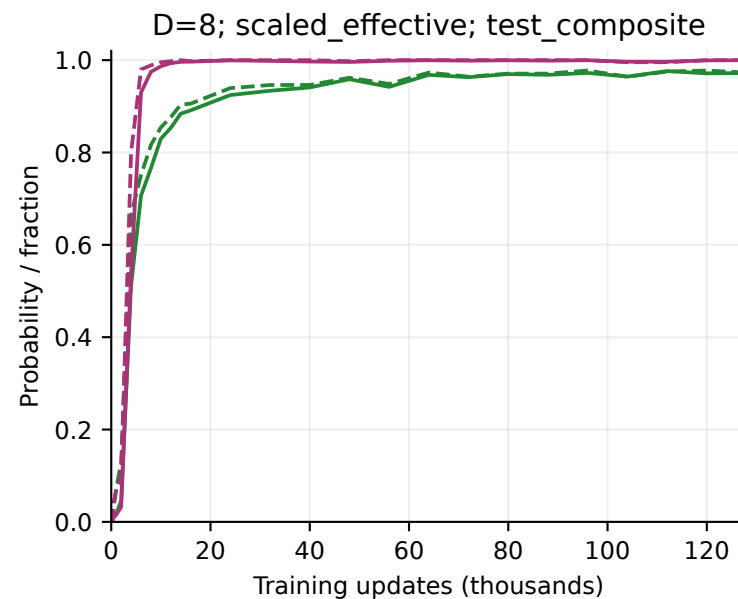
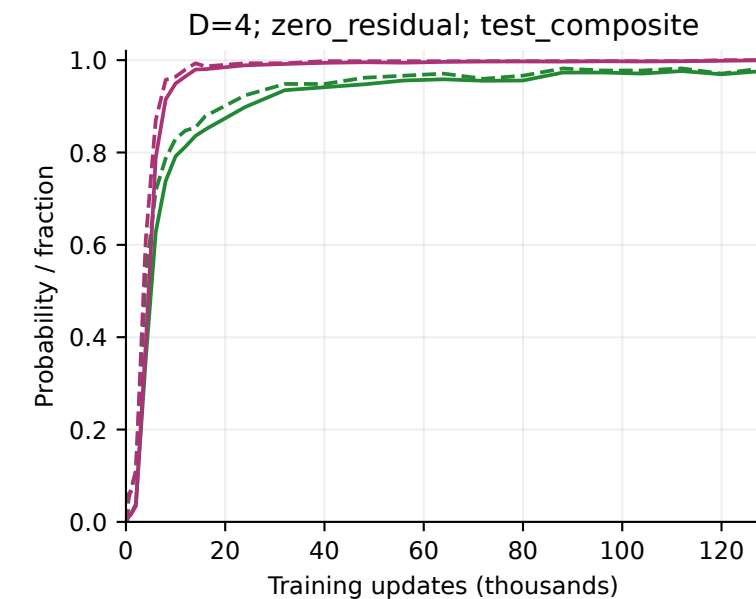
Confirmation OOD: all 3 worlds  $\times$  2 initializations; no checkpoint selection



Development: all 15 original runs; one world and initialization



Sensitivity: all 6 added runs; independent from formal matrix



— C1    — CD    — L2    - - - Answer accuracy  
— C2    — L1    — Mean P(gold answer)